

MATH 55 FINAL EXAM, PROF. SRIVASTAVA  
FRIDAY, MAY 13, 2016, 7:10–10:00PM, 100 LEWIS HALL.

Name: \_\_\_\_\_

SID: \_\_\_\_\_

INSTRUCTIONS: Write all answers in the provided space. Please write carefully and clearly, in complete English sentences. You may leave answers “unsimplified”, as sums/products/ratios possibly involving binomial coefficients. This exam includes three pages of scratch paper, which must be submitted, but will not be graded. Do not under any circumstances unstaple the exam. Write your name and SID on every page.

You are allowed to bring one letter-size single sided page of notes. You are not allowed to use any other materials or electronic devices.

UC BERKELEY HONOR CODE: *As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Sign here: \_\_\_\_\_

All question are ten points, except for 4,5,6 which are worth 8,6,6 points, respectively.

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| Do not turn over this page until your instructor tells you to do so. |
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Name and SID: \_\_\_\_\_

1. Circle true (**T**) or false (**F**) for each of the following. There is no need to provide an explanation.

(a) (2 points) The compound proposition

$$(p \rightarrow (q \wedge \neg q)) \rightarrow \neg p$$

is a tautology.

**T F**

(b) (2 points) There exist integers  $x$  and  $y$  such that  $21x + 54y = 1$ .

**T F**

(c) (2 points) A graph is 3-colorable if and only if all of the circuits in it have length divisible by 3.

**T F**

(d) (2 points) If  $E$  and  $F$  are events in a probability space such that  $p(E) = 0.99$  and  $p(F) = 0.99$  then  $E$  and  $F$  cannot be independent.

**T F**

(e) (2 points) Let  $I_n$  denote the number of injective functions from  $\{1, 2, 3, \dots, n\}$  to  $\{1, 2, \dots, 55\}$ . If  $m \geq n$  then it must be the case that  $I_m \geq I_n$ .

**T F**

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2. (10 points) Suppose  $A$  is a (possibly infinite) set. Prove that  $A$  and its power set  $\mathcal{P}(A)$  cannot have the same cardinality. (hint: Prove that there is no bijection  $f : A \rightarrow \mathcal{P}(A)$  by considering for any function  $f$  the subset  $B = \{x \in A : x \notin f(x)\}$ .)

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[Scratch Paper 1]

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3. Consider the following one player game. The game begins with  $n$  shapes, each of which is either a circle or a square. In each move, I have two options: (A) choose two shapes of the same type and replace them by a square, OR (B) choose two shapes of different types and replace them by a circle. The game is over when there is exactly one shape left, and I win if that shape is a circle (otherwise I lose).

(a) (4 points) Prove by induction that the game always terminates in a finite number of moves (regardless of the initial state).

(b) (6 points) Prove that I win the game if and only if the initial number of circles is odd.

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4. (8 points) For any prime  $p$ , consider the following relation on  $S = \{1, \dots, p-1\}$ :

$$R_p = \{(a, b) : a = b \vee ab \equiv 1 \pmod{p}\} \subseteq S \times S.$$

Prove or disprove: For every prime  $p$ ,  $R_p$  is an equivalence relation.

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5. (6 points) Calculate the remainder  $(-55)^{2016} \bmod 13$ .

6. (6 points) Suppose  $k \geq 1$  and  $(x_1, \dots, x_k)$  is a randomly chosen  $k$ -permutation of  $\{1, \dots, n\}$  (i.e., an ordered arrangement of  $k$  distinct elements, chosen uniformly from all such arrangements). What is the probability that it is a strictly increasing sequence, i.e., that

$$x_1 < x_2 < \dots < x_k?$$

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[Scratch Paper 2]



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7. (10 points) Consider a dial having a pointer that is equally likely to point to each of  $n$  regions numbered  $1, 2, \dots, n$ . If I spin the dial 3 times, what is the probability that the sum of the selected numbers is exactly  $n$ ?

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8. Suppose I roll a standard 6-faced die 7 times. Let  $X$  be a random variable equal to the number of **distinct** faces that appear (a face is a number in  $\{1, 2, 3, 4, 5, 6\}$ .)
- (a) (6 points) What is the expectation of  $X$ ?

- (b) (4 points) What is the variance of  $X$ ?

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9. (10 points) Consider the sequence  $a_0, a_1, a_2, \dots$  defined by the formula  $a_n = 3^n + 4^{n+2}$ . Derive a closed form expression for the generating function

$$G(x) = a_0 + a_1x + a_2x^2 + \dots$$

of this sequence, i.e., find polynomials  $p(x)$  and  $q(x)$  such that  $G(x) = p(x)/q(x)$ .

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[Scratch Paper 3]

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10. Prove or disprove:

- (a) (5 points) There exists a simple graph with 6 vertices and 5 edges such that  $G$  has an Euler circuit.

- (b) (5 points) There exists a simple graph  $G$  with 6 vertices and 7 edges such that  $G$  has an Euler circuit.

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11. (10 points) Prove by induction on  $k$  that every simple graph with  $n$  vertices and  $k$  edges has at least  $n - k$  connected components.